

Financial Analysis and Decision Making (FADM) Global BBA Fall 2026

Jason Kang

Session 1: Introduction and Time Value of Money

Jason Kang

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Grading and Key Dates

- ▶ In-Class Test (Oct 1, ~ 1 hour): **15%**
 - * Midterm Exam (Oct 12-16): **30%**
- ▶ Portfolio in Practice: In-Class Exam (Nov 19, ~ 1 hour): **15%**
 - * Final Exam (Dec 3-11): **40%**

If you miss **one** of the In-Class Test or the PiP exam, final exam = **55%**

If you miss **both**, you will get **0%**

Arrive on time.

Don't cheat Edusign (automatic F). I will count # in classroom.

Electronic devices. No laptops, except for Excel exercises in Week 3. Tablets are allowed only for note taking.
No cellphones outside of in-class problems (calculator).

General principle. Don't be a distraction.

Stop me if you need me to repeat.

Ask questions if you don't understand.

Participate in class discussions, and be respectful towards others.

- * The content of the class is identical to the material uploaded on K2, but the slides will slightly differ.
- * Real-world applications that are not part of the exam will be noted.

Plan For Today

- ▶ Course Logistics
- ▶ Overview of FADM
- ▶ Time Value of Money
- ▶ Intro to Perpetuities

SpaceX

+ Add to myFT

SpaceX sell-off wipes \$1tn from Elon Musk's rocket group

Stock trades below \$135 for first time since AI and space conglomerate's Wall Street debut in June

SpaceX stock dips below \$135 IPO price



Source: LSEG



“The subsequent slump in SpaceX’s stock price has come as shares in richly valued **tech stocks** have struggled over the past month over worries about **higher US interest rates** and the profitability of Big Tech’s **huge AI investments**.”

SpaceX sell-off wipes \$1tn from Elon Musk’s rocket group, Financial Times, 2026

Points raised:

- ▶ There is some relationship between interest rates and stock prices
- ▶ Tech stocks appear(?) to be affected more
- ▶ The profitability of investments (today) are also affected by interest rates

Example:

Suppose you have €10 million to invest in a company.

- ▶ **Option A:** SpaceX wants to use this money to invest in a new rocket launcher today. It expects to deliver profits (free cash flows) of €2mn starting from 10 years from now until forever.
- ▶ **Option B:** LVMH wants to use this money to build factories and expand production and train artisans to sell more handbags. It expects to deliver profits (free cash flows) of €2mn starting from 10 years from now until forever.

Which would you invest in?

Example:

Suppose you have €10 million to invest in a company.

- ▶ **Option A:** SpaceX wants to use this money to invest in a new rocket launcher today. It expects to deliver profits (free cash flows) of €2mn starting from 10 years from now until forever.
- ▶ **Option B (revised):** LVMH wants to use this money to build factories and expand production and train artisans to sell more handbags. It expects to deliver profits (free cash flows) of €0.5mn starting from 5 years from now until forever.

Which would you invest in?

FADM is an introductory course to finance and its first principles.

In FADM, our goal is to make “good decisions”

We will focus mostly on risk and discount rates.

Plan For Today

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Consider the following problem

$$€100 + \$100 = ?$$

Cannot add currencies without first converting into common currency

$$€100 + \$100 \times (0.86€/\$) = €186$$

$$€100 \times (\$1.16/€) + \$100 = \$216$$

Would you rather receive €100 now or €100 in one year?

- ▶ **€100** now because I can consume it immediately.
Immediate consumption is preferred to deferred consumption.
= impatience (preferences)
- ▶ **€100** now because I can grow it in one year.
I can deposit it at a bank at, for example, 5% interest rate, or invest in the stock market.
= opportunity cost
- ▶ **€100** now because I might not receive **€100** next year for sure.
The future is risky.
= risk

A euro today is not the same as a euro in one year.

- ▶ The discount rate is an exchange rate that equalizes the value of money at different points in time.
- ▶ Finance is the valuation of cash flows that accrue in the future.

Today, we are going to use r to denote the discount rate / interest rate.

For now, we are going to ignore risk entirely.

Suppose the interest rate is 5%. What is the value of €100 in one year if you deposit it in a bank?

For a single cash flow:

$$\text{Future Value (FV)} = C \times (1 + r)^t$$

where

- ▶ C = cash flow
- ▶ r = discount rate
- ▶ t = time

Suppose the interest rate is 5%. You need €105 in one year.
How much money do you deposit in a bank **today** to make sure you have €105 one year from now?

For a single cash flow:

$$\text{Present Value (PV)} = \frac{C}{(1 + r)^t}$$

Searching for r

What is the discount rate that makes you indifferent between receiving €100 today versus €105 in one year?

(Slide 14 in FADM Session 1 part 1)

Suppose $r = 7\%$. You need to buy a copier that costs \$9,500.

You can either buy it today, or buy it next year at \$10,000.

Which do you prefer?

Comparing Costs at Different Points in Time (slide 15 in FADM)

The cost of renovating a bridge was \$3bn in 2004. At the time, engineers estimated that if the project were delayed to 2005, the cost would rise by 10%. Assume $r = 2\%$. What is the cost of delay in terms of dollars in 2004?

Hint:

- Find the cost of doing the project in 2005
- Bring this cost to present value
- Compare this to the cost of doing the project in 2004.

Comparing Costs at Different Points in Time (Slide 15 in FADM Session 1 part 1)

The cost of renovating a bridge was \$3bn in 2004. At the time, engineers estimated that if the project were delayed to 2005, the cost would rise by 10%. Assume $r = 2\%$. What is the cost of delay in terms of dollars in 2004?

(Slide 20 in FADM Session 1 part 1)

Your firm has a risk-free investment opportunity where it can invest \$160,000 today and receive \$170,000 in one year. For what level of interest rates is this project attractive?

Usually, investment projects require an upfront cost in order to generate cash flows in the future.

- ▶ Develop a new drug (Sanofi)
- ▶ Build an electric vehicle (Renault)
- ▶ Create sustainable fuel (Airbus)

How does a CEO know whether to invest in a new project?

NPV Rule: Invest if the PV of benefits $>$ PV of costs

Definition. $NPV = PV \text{ of benefits} - PV \text{ of costs}$

NPV Rule. Invest if $NPV > 0$

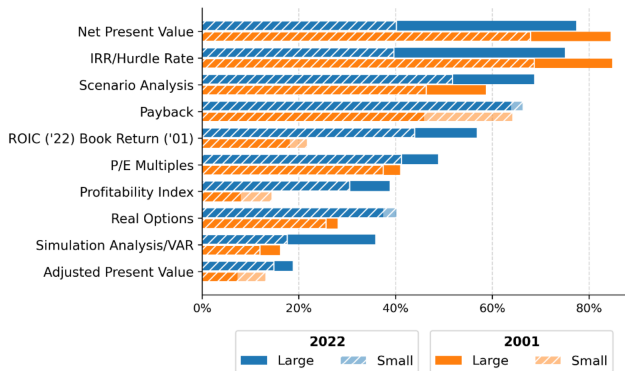
Example.

Assume $r = 10\%$. Invest in an oven today for €80. Sell croissants next year and make €110. $NPV = ?$

Concept: Net Present Value

Figure 2. Capital Budgeting Techniques

This figure displays CFO responses to the following question about the capital budgeting methods their companies use to make investment decisions: *How frequently does your firm use the following techniques when deciding which projects or acquisitions to pursue?* {0 = Never, 1, 2, 3, 4 = Always}. The percentage of firms that answer 3 or 4 are shown. The blue bars display results for the 2022 Duke CFO survey (March 2020 wave); the orange bars display results from Graham and Harvey (2001). Within each blue and orange bar, the solid portion displays responses for large firms (revenue above \$1 billion), and the white, crosshatched portion displays responses for small firms (revenue below \$1 billion). For example, 77% of large firms (40% of small firms) say that Net Present Value was an important or very important decision rule in 2022.



(Slide 27 in FADM Session 1 part 1)

A project costs \$10mn today and \$5mn next year. Next year, it will generate \$20mn. Assume $r = 10\%$.

- a. What is the NPV of this project?

(Slide 27 in FADM Session 1 part 1)

A project costs \$10mn today and \$5mn next year. Next year, it will generate \$20mn. Assume $r = 10\%$.

- a. What is the NPV of this project?

Skip slides 28-29 (part b)

(Slide 32 in FADM Session 1 part 1)

You need to purchase 10 keyboards from a supplier.

- ▷ Supplier A asks you for \$100 today plus \$10 per keyboard one year from now
- ▷ Supplier B wants \$21 per keyboard one year from now

Assume $r = 6\%$. Which supplier is cheaper?

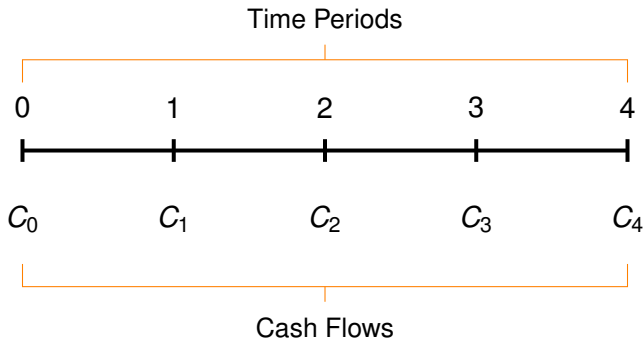
(Slide 32 in FADM Session 1 part 1)

You need to purchase 10 keyboards from a supplier.

- ▷ Supplier A asks you for \$100 today plus \$10 per keyboard one year from now
- ▷ Supplier B wants \$21 per keyboard one year from now

Assume $r = 6\%$. What is the total cost in the perspective of $t = 1$? (Skip slide 34.)

Tools for Multiple Cash Flows

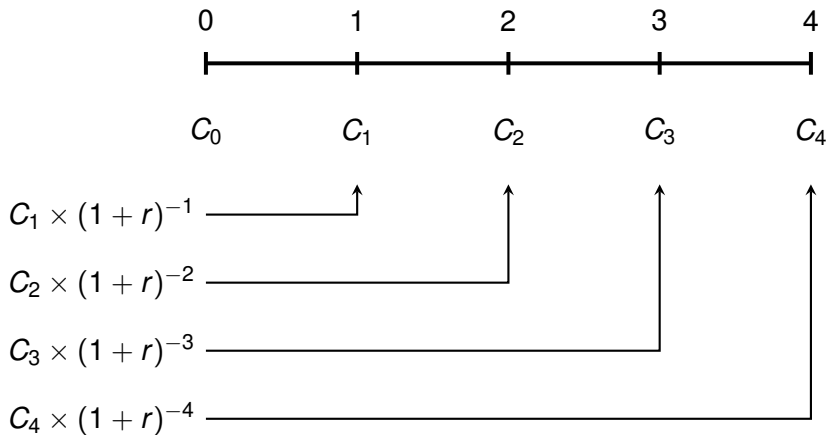


Convention: treat each time period as at the end of a full year.

Always draw a timeline to visualize the timing of cashflows!

The value of C today ($= PV_0$)

The value of C today ($= PV_0$)



$$PV = \sum_t \frac{C_t}{(1+r)^t} = C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \dots$$

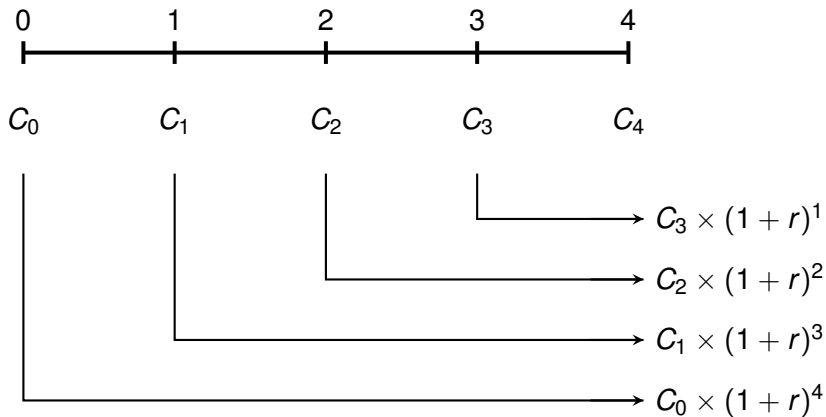
The value of C today ($= PV_0$)

(Slide 43 in FADM Session 1 part 1)

Assume $r = 3.5\%$. You will receive \$10 in 1 year, \$2 in 2 years and \$3 in 3 years.

- a. What is the PV of this stream of cash flows?

The value of C in 4 years ($= FV_4$)



We can add these cash flows because they are in same (future) time unit

The value of C in 4 years ($= FV_4$)

(Slide 43 in FADM Session 1 part 1)

Assume $r = 3.5\%$. You will receive \$10 in 1 year, \$2 in 2 years and \$3 in 3 years.

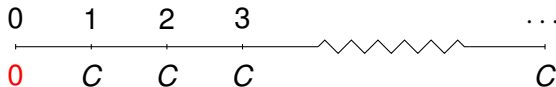
- b. What is the future value of this stream of cash flows at the end of year 3?

Recap: Time Value of Money

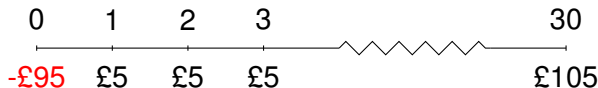
Plan For Today

- ▶ Course Logistics
- ▶ Overview of FADM
- ▶ Time Value of Money
- ▶ **Intro to Perpetuities**

Definition. A perpetuity is a stream of fixed payments that last forever. By convention, the date of the first payment is at $t = 1$.



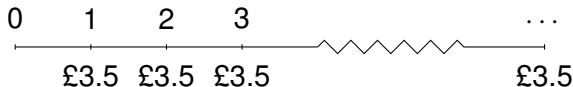
In 1917, the British government issued war bonds with a principal of £100 to be repaid in 30 years, in addition to annual payments of £5. Each bond was purchased at £95, leading to an effective interest rate of 5.25%.



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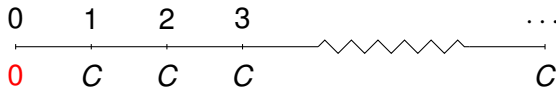
Perpetuity

In 1932, the UK govt converted the now-17 year bond to a perpetual (consol) bond, now paying £3.5 forever.



If an investor were to sell this bond, what would its value be?

Definition. A perpetuity is a stream of fixed payments that last forever. By convention, the date of the first payment is at $t = 1$.



We can represent the present value of this perpetuity as

$$PV_0 = \sum_{t=1}^{\infty} \frac{C}{(1+r)^t}$$

- ▶ According to *Wikipedia* under “geometric series”, the series

$$\sum_{k=0}^{\infty} a \cdot z^k$$

converges to $\frac{a}{1-z}$ if and only if $|z| < 1$.

- ▶ We have something similar!

$$PV_0 = \sum_{t=1}^{\infty} C \cdot \left(\frac{1}{1+r} \right)^t$$

- Add and subtract,

$$\begin{aligned}PV_0 &= C + \sum_{t=1}^{\infty} C \cdot \left(\frac{1}{1+r}\right)^t - C = \sum_{t=0}^{\infty} C \cdot \left(\frac{1}{1+r}\right)^t - C \\&= \frac{C}{1 - \left(\frac{1}{1+r}\right)} - C = \frac{\left(\frac{1+r}{1+r}\right) C}{\frac{1+r}{1+r} - \frac{1}{1+r}} - C = \frac{(1+r) \cdot C}{r} - C \\&= \boxed{\frac{C}{r}}\end{aligned}$$

- Main takeaway: if first cash flow starts at 0:

$$PV_0 = \sum_{t=0}^{\infty} C \cdot \left(\frac{1}{1+r} \right)^t = C + \frac{C}{r}$$

- If first cash flow starts at 1:

$$PV_0 = \sum_{t=1}^{\infty} C \cdot \left(\frac{1}{1+r} \right)^t = \frac{C}{r}$$

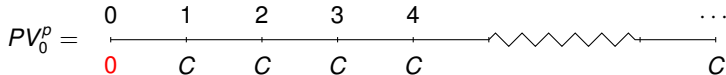
Example

Assume $r = 2\%$. What is the NPV of a project that requires an initial investment of €100 today but gives you €10 each year (starting from next year)?

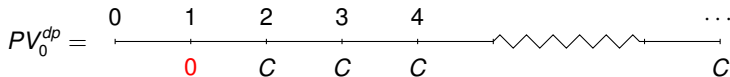
► Deferred perpetuity

What if the first cash flow is delivered n periods **after** time 1?

– Normal case: perpetuity with first payment at 1:



– Deferred perpetuity with first payment at 2:



► We lost one payment of $\frac{C}{1+r}$.

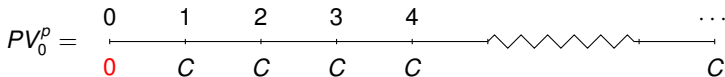
$$PV_0^{dp} = PV_0^p - \frac{C}{1+r} = \frac{C}{r} - \frac{C}{1+r} = \frac{C(1+r)}{r(1+r)} - \frac{C \cdot r}{r(1+r)} = \frac{C}{r} \left(\frac{1}{1+r} \right)$$

Deferred perpetuity

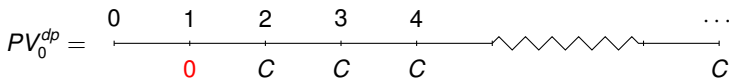
► Deferred perpetuity

What if the first cash flow is delivered n periods **after** time 1?

– Normal case: perpetuity with first payment at 1:



– Deferred perpetuity with first payment at 2:



- Or, a deferred perpetuity today is just a perpetuity tomorrow, so 'translate' back in time,

$$PV_0^{dp} = \left(\frac{1}{1+r} \right) \times PV_1^p = \left(\frac{1}{1+r} \right) \frac{C}{r}$$

Deferred perpetuity

Deferred perpetuity

In general,
$$PV_0 = \frac{C}{r} \left(\frac{1}{1+r} \right)^{t-1}$$

where t is the date of the first payment.

i. What if $t = 0$?

Deferred perpetuity

In general,
$$PV_0 = \frac{C}{r} \left(\frac{1}{1+r} \right)^{t-1}$$

where t is the date of the first payment.

ii. What if $t = 1$?

Deferred perpetuity

In general,
$$PV_0 = \frac{C}{r} \left(\frac{1}{1+r} \right)^{t-1}$$

where t is the date of the first payment.

iii. What if $t = 2$?

(Slide 9 in FADM Session 1 part 2)

Assume $r = 9.5\%$. You pay €100 today to build a machine that gives you €10 each year forever, starting from next year. What is the NPV of this project?

(Slide 10 in FADM Session 1 part 2)

Assume $r = 9.5\%$. You pay €100 today to build a machine that gives you €10 each year forever. However, it takes one additional year to construct. What is the NPV of this project?

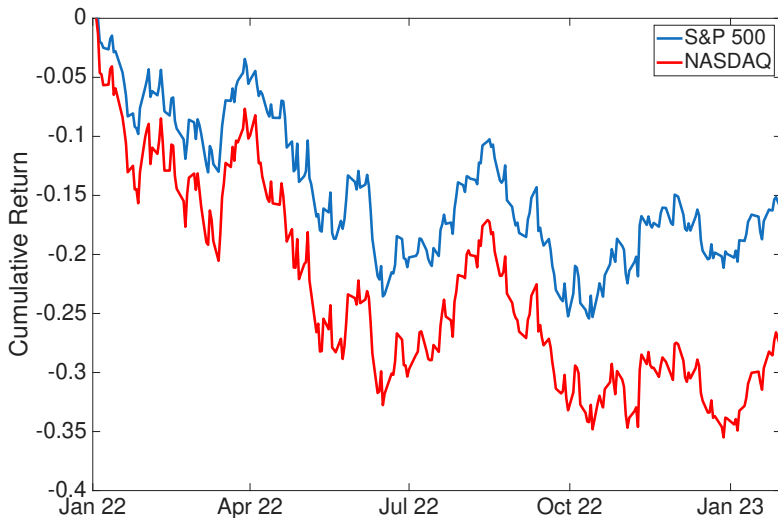
Do higher interest rates hurt growth stocks?



“The standard explanation for this ... is that higher rates mean a higher discount rate applied to future cash flows; so big cash flows far into the future are worth relatively less when rates rise; so growth stocks, which carry high expectations for distant cash flows, get hit harder.”

Financial Times, 2022

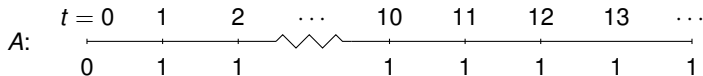
Do higher interest rates hurt growth stocks?



Example

Suppose there are two investments, A and B . Starting next year, A gives you \$1 annually forever. On the other hand, B gives you \$2 at perpetuity, but starting 10 years from now. Assume $r = 2\%$.

a. What is the value of investment A ?

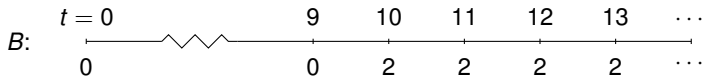


► **Q** : What do the cash flows for A look like?

Example

Suppose there are two investments, A and B . Starting next year, A gives you \$1 annually forever. On the other hand, B gives you \$2 at perpetuity, but starting 10 years from now. Assume $r = 2\%$.

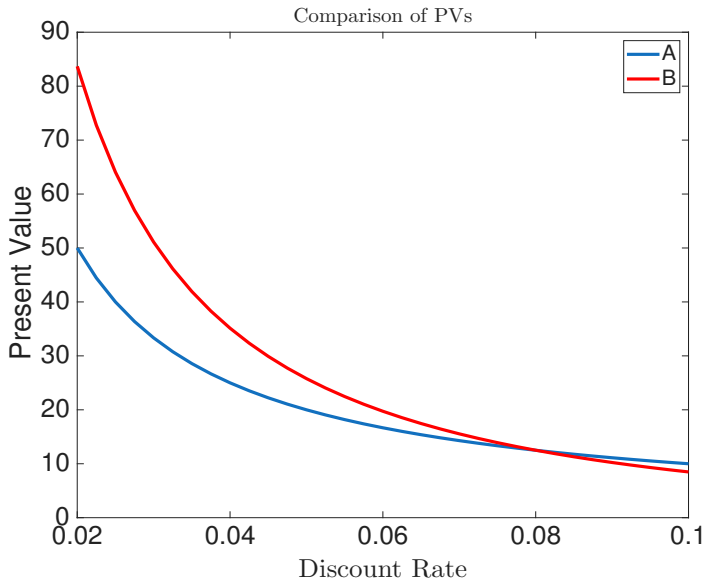
b. What is the value of investment B ?



► **Q** : What do the cash flows for B look like? **A** : A deferred perpetuity.

Example

Example



- Present Value

$$PV_0 = \frac{C}{(1+r)^t}$$

- Future Value

$$FV_t = C \times (1+r)^t$$

- Perpetuity

$$PV_0 = \frac{C}{r}$$

- Deferred Perpetuity

$$PV_0 = \frac{C}{r} \left(\frac{1}{1+r} \right)^{t-1}$$

where t is the date of the first cash flow

